

README.md

KNPC Market Intelligence System Master Overview

The KNPC Market Intelligence System is an enterprise Business Intelligence (BI) dashboard and data orchestration environment built to track, evaluate, and report global crude oil benchmarks and regional refined product proxy values. The platform isolates ingestion tasks from downstream presentation dashboards, logs execution states, handles network data parsing issues, and compiles operational management workbooks.

SYSTEM LAYOUT SEQUENCE MAPPING

```
Public Web Endpoints (KPC, Yahoo Finance, OilPrice)
|
v HTTP Request Scraping Loops
`collectors.py` Ingestion Module
|
v Relational Structured Database Writes
SQLite Engine Store (`market_data.db`)
|
v Processing & OpenPyXL File Compilations
`analytics.py` & `exporter.py` Suites
|
v Dashboard Layout Rendering UI
`app.py` Streamlit Application Node
```

REPOSITORY BLUEPRINT REGISTRY

- app.py: Core UI built over Streamlit. Controls login verification grids, metrics rendering frames, charts, and data adjustment widgets.
- main.py: Execution entry node. Initializes SQLite structural components, handles script schedules, and builds export sheets.
- collectors.py: Extraction engine layer utilizing explicit regular expressions and user-agent manipulation loops.
- analytics.py: Processing layer calculating temporal metrics adjustments, data splits, and tracking spreads.
- database.py: Controls backend connection paths, asset structural layouts, and execution logs.
- exporter.py: Constructs automated tabular spreadsheets and processing summaries split across asset sheets.

Deploy.md

Platform Provisioning, Configuration, and Isolation Procedures

HOST PREREQUISITES

Verify that target execution servers host a clean Python v3.10-v3.12 stack. Ensure external network firewall rule routes allow outward HTTPS operations over Port 443 to reach target data locations: eapp.kpc.com.kw, query1.finance.yahoo.com, and oilprice.com.

ENVIRONMENT VARIABLE SETUPS (.ENV)

```
# Define active rolling matrix evaluation lookback window
ANALYTICS_LOOKBACK_DAYS=7
```

MANUAL SETUP INSTRUCTION MAPPING

```
git clone <repository_url> market-intel && cd market-intel
python -m venv venv
source venv/bin/activate
pip install -r requirements.txt
python main.py
streamlit run app.py --server.port 8501 --server.address 0.0.0.0
```

User_Manual.md

Operations Dashboard Interface and Compliance Adjustment Workflows

1. SECURITY PORTAL ACCESS GATE

Accessing the dashboard web link opens an authentication window. Use the default administrative configurations:

- Username: admin
- Password: admin

If inputs do not match standard clearance requirements, structural evaluation frameworks block page generation logs.

2. CENTRAL DASHBOARD FUNCTIONAL MATRICES

- Crude Oil Global Benchmarks: Formats interactive indicators tracking pricing sets for Brent, WTI, Oman, and Kuwait Export Crude.
- Regional Product Price Proxies: Tracks refined targets (Naphtha, Gasoil, Gasoline 92/95, Jet Fuel, Bunker Oil) against transparent regional points when key numbers are paywalled.
- Data Adjustment Ledger: Enables operators to review collection records and directly write missing numbers when primary web systems fail to update.

Developer_Manual.md

Database Structural Schemas and Resilient Failover Logics

1. CORE DATABASE ARCHITECTURE (MARKET_DATA.DB)

The relational storage structure isolates operational historical traces into dedicated tracking objects:

- crude_prices: Stores raw extracted data fields. Index tracking is governed via: UNIQUE(price_date, benchmark, source).
- product_prices: Contains fields for market tracking, product proxy configurations, status, and manual annotations.
- product_collection_attempts: Keeps an exact trace of scraper network exceptions, timeouts, and tracking variations.

2. INGESTION ERROR MITIGATION AND BADGING RULES

If an absolute parsing failure occurs or a page structure shifts during a scraping sequence, the collectors.py module protects overall dashboard stability by performing a fallback data transaction:

- Price Column Value: Writes an explicit SQL NULL row.
- Source Mapping Field: Records 'Source system didn't publish'.
- Data Processing Status: Labels the record as 'manual_required'.

The processing engine filters out these elements from moving charts, while app.py flags the missing field with a high-visibility warning badge, signaling to the refinery coordinator that a manual data override is needed.